

AbangCebu AI “ Sprint 2 Master Plan & Execution Specification (Granular Module Breakdown) **Document Version:** 2.0.0 **Status:** Approved for Sprint 2 Execution **Project:** AbangCebu AI (`abang-cebu-ai`) **Target Sprint:** Sprint 2 (Sprint ID: `70` / Board ID: `1`) **Architecture Granularity:** Option B (Granular Feature/Screen-Level Modules) **Author:** Hermar Centillas (Lead / Scrum Master) **Co-Authors:** Engineering Team & QA Lead (Jenny Villamor) **Created Date:** October 1, 2026 --- ## 1. Executive Summary & Granularity Rationale In response to academic and project review requirements specifying a **strict per-module breakdown**, the system architecture is decomposed into **16 granular, self-contained functional modules** (Option B). Under this model: - Sub-processes previously grouped under parent umbrellas (such as Login, Registration, and Password Reset) are formalized as **independent system modules** with their own workflows, actors, database interactions, error states, and dedicated `.drawio` flowchart specifications. - Features with dual interfaces (such as Interactive Map Discovery vs Structured Attribute Filter Search) are cleanly decoupled to provide modular clarity for developers, testers, and academic evaluators. --- ## 2. Section A: Final Granular Module List (16 Modules) | # | Module Name | System Description & Scope | Current Status | Primary Evidence / Source |

---	---	---	---	---	---		**1**	**User Registration Module**	Renter and Landlord account creation, role selection, NIST SP 800-63B password complexity, Philippine mobile format normalization (+639XXXXXXXX), Turnstile bot validation, and PKCE verification email dispatch.	**Foundation Implemented / Pre-UI**	[`src/types/auth.ts`](file:///home/hrmr/abang-cebu-ai/src/types/auth.ts), [`docs/specifications/auth/auth-registration-spec.md`](file:///home/hrmr/abang-cebu-ai/docs/specifications/auth/auth-registration-spec.md), [`docs/design/registration-page-wireframe.md`](file:///home/hrmr/abang-cebu-ai/docs/design/registration-page-wireframe.md)
							2	**User Login & Session Module**	User authentication, Supabase GoTrue JWT exchange, HTTP-only cookie chunking (sb*-auth-token), Refresh Token Rotation (RTR), multi-tab synchronization (BroadcastChannel), and role-based route redirects.	**Foundation Implemented / Pre-UI**	[`src/types/auth.ts`](file:///home/hrmr/abang-cebu-ai/src/types/auth.ts), [`middleware.ts`](file:///home/hrmr/abang-cebu-ai/middleware.ts), [`docs/specifications/auth/auth-session-lifecycle.md`](file:///home/hrmr/abang-cebu-ai/docs/specifications/auth/auth-session-lifecycle.md), [`docs/design/login-page-wireframe.md`](file:///home/hrmr/abang-cebu-ai/docs/design/login-page-wireframe.md)
							3	**Password Reset & Recovery Module**	Self-service password recovery, anti-enumeration security timing, single-use PKCE recovery email dispatch, password updates, and mandatory global revocation of older sessions (scope: 'global').	**Foundation Implemented / Pre-UI**	[`docs/specifications/auth/auth-password-reset-spec.md`](file:///home/hrmr/abang-cebu-ai/docs/specifications/auth/auth-password-reset-spec.md), [`docs/testing/auth-test-plan.md`](file:///home/hrmr/abang-cebu-ai/docs/testing/auth-test-plan.md)
							4	**User Profile Management Module**	Management of `public.profiles`, personal information updates (full name, phone number, bio), avatar image		

uploads to Supabase storage with MIME/size limits, and account preferences. |
Foundation Implemented / In Development | [docs/database/users-and-profiles-schema.md](file:///home/hrmr/abang-cebu-ai/docs/database/users-and-profiles-schema.md), [src/types/database.ts](file:///home/hrmr/abang-cebu-ai/src/types/database.ts), [docs/design/profile-page-wireframe.md](file:///home/hrmr/abang-cebu-ai/docs/design/profile-page-wireframe.md) | | **5** | **Property Listing Creation Module** | Landlord property creation wizard: property metadata (building type, address, barangay, city, PostGIS coordinates), multi-unit configurations (rent, deposit, rooms, rules), amenity tagging, and photo uploads. |
Schema Defined / Planned | [docs/database/database-erd.md](file:///home/hrmr/abang-cebu-ai/docs/database/database-erd.md) (`properties`, `rental\_units`, `property\_photos`, `amenities`), [docs/architecture/what-is-abangcebu-ai.md](file:///home/hrmr/abang-cebu-ai/docs/architecture/what-is-abangcebu-ai.md) | |
6 | **Unit Availability & Management Module** | Landlord listing dashboard: toggling real-time room/bedspace occupancy (`is\_available`), adjusting monthly rental rates, archiving units, and reviewing unit-specific views and inquiries. |
Schema Defined / Planned | [docs/database/database-erd.md](file:///home/hrmr/abang-cebu-ai/docs/database/database-erd.md) (`rental\_units.is\_available`), [docs/design/user-dashboard-wireframe.md](file:///home/hrmr/abang-cebu-ai/docs/design/user-dashboard-wireframe.md) | | **7** | **Interactive Map Discovery Module** | Geospatial rental visualization across Metro Cebu using MapLibre GL: viewport bounding-box synchronization (`&& ST\_MakeEnvelope`), interactive price pins, geolocation centering, and pin selection preview drawers. |
Schema Defined / Planned | [docs/database/database-erd.md](file:///home/hrmr/abang-cebu-ai/docs/database/database-erd.md) (PostGIS section), [docs/design/mobile-map-wireframes.md](file:///home/hrmr/abang-cebu-ai/docs/design/mobile-map-wireframes.md), [src/components/map/](file:///home/hrmr/abang-cebu-ai/src/components/map/) | | **8** | **Property Search & Filter Module** | Parametric multi-facet search engine: filtering by budget min/max, room category (bedspace, room, condo), gender rules, curfew, visitor policy, utilities, and landmark radius (`ST\_DWithin`). |
Schema Defined / Planned | [docs/architecture/what-is-abangcebu-ai.md](file:///home/hrmr/abang-cebu-ai/docs/architecture/what-is-abangcebu-ai.md), [docs/database/database-erd.md](file:///home/hrmr/abang-cebu-ai/docs/database/database-erd.md) | | **9** | **AI Rental Assistant Module** | Natural language conversational finder supporting English and Cebuano/Bisaya; translates prompts into structured query filters; resolves ambiguous landmarks; strictly grounded in approved database listings. |
Behavioral Spec Defined / Planned | [docs/architecture/what-is-abangcebu-ai.md](file:///home/hrmr/abang-cebu-ai/docs/architecture/what-is-abangcebu-ai.md) (Sections: "The Role of AI", "How It Works") | | **10** | **Inquiry & Viewing Reservation Module** | Manages renter inquiries on properties and schedules in-person ocular viewing visits (`viewing\_date`); handles landlord acceptance/rescheduling state machine. Excludes financial payment processing. |
Schema Defined / Planned | [docs/database/database-erd.md](file:///home/hrmr/abang-cebu-ai/docs/database/database-erd.md)

hrmr/abang-cebu-ai/docs/database/database-erd.md) (`inquiries` table), [docs/architecture/what-is-abangcebu-ai.md](file:///home/hrmr/abang-cebu-ai/docs/architecture/what-is-abangcebu-ai.md) | | ****11**** | ****Landlord KYC & Verification Module**** | Anti-scam trust layer: Landlord government ID upload (PhilSys, Passport, PRC, UMID) and proof of property ownership; private storage RLS; administrative review queue before public listing publishing. | ****Schema & Types Defined / Planned**** | [docs/database/database-erd.md](file:///home/hrmr/abang-cebu-ai/docs/database/database-erd.md) (`kyc\_verifications`), [src/types/database.ts](file:///home/hrmr/abang-cebu-ai/src/types/database.ts) (`KycStatus`, `KycIdType`) | | ****12**** | ****Messaging & Contact Module**** | Direct communication channel between renters and verified landlords; exposes verified phone numbers, direct contact links, and inquiry conversation threads. | ****Architectural Scope Defined / Planned**** | [docs/architecture/what-is-abangcebu-ai.md](file:///home/hrmr/abang-cebu-ai/docs/architecture/what-is-abangcebu-ai.md), [docs/database/database-erd.md](file:///home/hrmr/abang-cebu-ai/docs/database/database-erd.md) | | ****13**** | ****Review & Rating Module**** | Verified tenant evaluation workflow: 1-to-5 star ratings, qualitative feedback, verified tenant badges, and aggregate rating score calculations on properties. | ****Schema Defined / Planned**** | [docs/database/database-erd.md](file:///home/hrmr/abang-cebu-ai/docs/database/database-erd.md) (`reviews` table) | | ****14**** | ****Reporting & Scam Flagging Module**** | Community moderation tool: allows users to flag fraudulent listings, inaccurate pricing, or abusive landlords; triggers immediate administrative alerts in audit queue. | ****Schema Defined / Planned**** | [docs/database/database-erd.md](file:///home/hrmr/abang-cebu-ai/docs/database/database-erd.md) (`audit\_logs`), [docs/architecture/what-is-abangcebu-ai.md](file:///home/hrmr/abang-cebu-ai/docs/architecture/what-is-abangcebu-ai.md) | | ****15**** | ****Admin Management & Moderation Module**** | Central governance portal for administrators: listing approval/rejection queue, landlord KYC verification review, account suspensions (`is\_suspended`), and immutable audit log inspection. | ****Wireframe & Schema Defined / Planned**** | [docs/design/admin-dashboard-wireframe.md](file:///home/hrmr/abang-cebu-ai/docs/design/admin-dashboard-wireframe.md), [docs/database/database-erd.md](file:///home/hrmr/abang-cebu-ai/docs/database/database-erd.md) (`audit\_logs`), [src/types/database.ts](file:///home/hrmr/abang-cebu-ai/src/types/database.ts) (`is\_admin`) | | ****16**** | ****Notification Module**** | Transactional alerting system: dispatches email and in-app badge notifications for inquiry status transitions, viewing confirmations, KYC verification outcomes, and security alerts. | ****Architectural Scope Defined / Planned**** | [docs/specifications/auth/auth-registration-spec.md](file:///home/hrmr/abang-cebu-ai/docs/specifications/auth/auth-registration-spec.md), [docs/database/database-erd.md](file:///home/hrmr/abang-cebu-ai/docs/database/database-erd.md) | --- ## 3. Section B: Flowchart Deliverables Directory (docs/flowcharts/) Each of the 16 modules receives its own dedicated `.drawio` file, adopting standard uncompressed XML and clean monochrome formatting: ``text docs/â”€â”€â”€â”€ flowcharts/â”€â”€â”€â”€ user-registration.drawio # Module 1:

Registration, Validation & PKCE Activation â”œâ”€â”€ user-login.drawio # Module
 2: Login, Session Token & Role Routing â”œâ”€â”€ password-reset.drawio #
 Module 3: Self-Service Password Recovery & Invalidation â”œâ”€â”€ user-profile-
 management.drawio # Module 4: Profile Details & Avatar Upload â”œâ”€â”€
 property-listing-creation.drawio # Module 5: Landlord Multi-Unit Listing
 Submission â”œâ”€â”€ unit-availability.drawio # Module 6: Real-Time Occupancy
 & Unit Controls â”œâ”€â”€ interactive-map.drawio # Module 7: MapLibre
 Viewport Panning & Spatial Query â”œâ”€â”€ property-search.drawio # Module 8:
 Multi-Facet Filter & Radius Search â”œâ”€â”€ ai-rental-assistant.drawio # Module
 9: Natural Language Parsing & Recommendations â”œâ”€â”€ inquiry-
 reservation.drawio # Module 10: Ocular Viewing Requests & Scheduling State
 â”œâ”€â”€ landlord-verification.drawio # Module 11: Government ID Submission
 & Trust Pipeline â”œâ”€â”€ messaging-contact.drawio # Module 12: Direct Inquiry
 Communication & Contact â”œâ”€â”€ review-rating.drawio # Module 13: Verified
 Tenant Ratings & Reviews â”œâ”€â”€ report-moderation.drawio # Module 14:
 Fraud Reporting & Moderation Flagging â”œâ”€â”€ admin-management.drawio #
 Module 15: Admin Governance, Moderation & Audit Logs â””â”€â”€
 notification.drawio # Module 16: Transactional Alerts & In-App Notices `` -- ## 4.
 Section C: Revised Sprint 2 Task Breakdown (Sprint 1 Standard Alignment) ``text
 SPRINT 2 TASK STRUCTURE (MODULES, FLOWCHARTS & TEST SPECIFICATIONS)
 â”œâ”€â”€ STREAM 1: System Module Architecture & Boundaries (SCRUM-75)
 â”œâ”€â”€ STREAM 2: System Module Flowchart Authoring & Visual Verification
 (SCRUM-76 to SCRUM-81) â”œâ”€â”€ STREAM 3: Modular QA Test Specifications &
 Test Case Migration (SCRUM-82 to SCRUM-87) â”œâ”€â”€ STREAM 4: Security RBAC
 & Middleware Route Test Matrices (SCRUM-88) â”œâ”€â”€ STREAM 5: Spatial & AI
 Performance Baseline Strategy (SCRUM-89) â””â”€â”€ STREAM 6: Master QA
 Workbook & Deliverable Sign-Off (SCRUM-90 to SCRUM-91) `` ### Stream 1:
 System Module Architecture & Boundaries - **SCRUM-75: Module Architecture
 Finalization & System Boundary Sign-Off** (Done / Lead) Finalize 16 granular
 modules across Identity, Marketplace, Discovery, AI, Trust, and Governance; sign
 off on zero in-app payment / escrow boundary. ### Stream 2: System Module
 Flowchart Authoring & Visual Verification - **SCRUM-76: Author Authentication &
 User Management Flowcharts (4 .drawio files)** (Done / John Lloyd Ando & Lead)
 Modules 01â”€04 (`user-registration.drawio`, `user-login.drawio`, `password-
 reset.drawio`, `user-profile-management.drawio`). - **SCRUM-77: Author Property
 Listing & Availability Flowcharts (2 .drawio files)** (Done / Dency Marie Bosque &
 Lead) Modules 05â”€06 (`property-listing-creation.drawio`, `unit-
 availability.drawio`). - **SCRUM-78: Author Interactive Map & Search Filter
 Flowcharts (2 .drawio files)** (Done / Angel Crushein Yaun, Neah Moneva & Lead)
 Modules 07â”€08 (`interactive-map.drawio`, `property-search.drawio`). -
 **SCRUM-79: Author AI Assistant & Viewing Reservation Flowcharts (2 .drawio
 files)** (Done / Michelle Estoy, Edrich Bardilas & Lead) Modules 09â”€10 (`ai-
 rental-assistant.drawio`, `inquiry-reservation.drawio`). - **SCRUM-80: Author
 Landlord KYC & Direct Messaging Flowcharts (2 .drawio files)** (Done / Michelle

Estoy, Edrich Bardilas & Lead) Modules 11â€“12 (`landlord-verification.drawio`, `messaging-contact.drawio`). - **SCRUM-81: Author Review, Moderation & Notification Flowcharts (4 .drawio files)** (Done / Anne KC, Karla, Ryza & Lead) Modules 13â€“16 (`review-rating.drawio`, `report-moderation.drawio`, `admin-management.drawio`, `notification.drawio`). ### Stream 3: Modular QA Test Specifications & Test Case Migration - **SCRUM-82: Author Auth & Identity Lifecycle QA Test Specification (Modules 01â€“04)** (1.5d / Jenny Villamor & John Lloyd) Migrate and expand 24 test cases (`TC-REG-01..07`, `TC-LOG-01..07`, `TC-RST-01..05`, `TC-PROF-01..05`) from Sprint 1 foundations. - **SCRUM-83: Author Property Listing & Availability QA Test Specification (Modules 05â€“06)** (1.5d / Dency Marie Bosque & Jenny Villamor) Author 10 test cases (`TC-PROP-01..06`, `TC-AVAIL-01..04`) covering multi-unit wizard, PostGIS coordinate validation, and occupancy toggles. - **SCRUM-84: Author Interactive Map & Parametric Search QA Test Specification (Modules 07â€“08)** (1.5d / Angel Crushein & Neah Moneva) Author 11 test cases (`TC-MAP-01..05`, `TC-SRCH-01..06`) covering viewport bounding box (`ST\_MakeEnvelope`), clustering, and landmark radius (`ST\_DWithin`). - **SCRUM-85: Author AI Rental Assistant & Viewing Reservation QA Test Specification (Modules 09â€“10)** (1.5d / Michelle Estoy & Edrich Bardilas) Author 10 test cases (`TC-AI-01..05`, `TC-INQ-01..05`) covering natural language Bisaya/English queries, grounding defenses, and ocular viewing scheduling. - **SCRUM-86: Author Landlord KYC Verification & Direct Messaging QA Test Specification (Modules 11â€“12)** (1.5d / Michelle Estoy & Edrich Bardilas) Author 9 test cases (`TC-KYC-01..05`, `TC-MSG-01..04`) covering PhilSys/Passport validation, private storage RLS, and direct verified messaging. - **SCRUM-87: Author Governance, Moderation & Notification QA Test Specification (Modules 13â€“16)** (1.5d / Anne KC, Karla & Ryza) Author 14 test cases (`TC-REV-01..04`, `TC-REP-01..04`, `TC-ADM-01..04`, `TC-NOTIF-01..02`) covering reviews, scam reports, admin console, and alerts. ### Stream 4: Security RBAC & Middleware Route Test Matrices - **SCRUM-88: Author Sprint 2 RBAC & Route Middleware Test Specification (All 16 Modules)** (1.0d / Karla Hiyas & Anne KC) Expand Sprint 1 RBAC and middleware test scenarios to govern route permissions across all 16 granular modules. ### Stream 5: Spatial & AI Performance Baseline Strategy - **SCRUM-89: Author Sprint 2 Spatial & AI Query Performance Baseline Strategy** (1.0d / Ryza Albiso) Formulate latency SLAs (<200ms PostGIS viewport, <1.5s AI streaming) mirroring Sprint 1 `auth-performance-baseline.md`. ### Stream 6: Master QA Workbook & Deliverable Sign-Off - **SCRUM-90: Compile Sprint 2 Master QA Test Specification Workbook (Excel .xlsx Companion)** (1.0d / Jenny Villamor) Generate multi-sheet workbook `AbangCebu\_Sprint2\_Master\_Test\_Specification.xlsx` adhering to `Test\_Specification\_Template.xlsx`. - **SCRUM-91: Compile Sprint 2 Master QA Test Plan & Audit Sign-Off Package (PDF)** (0.5d / Lead & Jenny Villamor) Compile `AbangCebu\_Sprint2\_Master\_Test\_Plan.pdf` and synchronize all Jira issues for Sprint 2 defense. --- ## 5. Section D: Master QA Test Plan (Option B Mappings â€“ 77 Test Cases) All 16 granular modules are mapped to corresponding test suites in

`docs/testing/sprint-2-master-test-plan.md` and `docs/testing/AbangCebu\_Sprint2\_Master\_Test\_Specification.xlsx`: ``

+-----+			
GRANULAR QA TEST SUITE TRACEABILITY MATRIX (77 CASES) +-----			
+-----+-----+-----+ Domain			
Granular Module Name	Covered QA Suite & Test Case IDs	Jira Ticket	
+-----+-----+-----+-----+			
Identity User Registration Module Suite 1: Registration (`TC-REG-01..07`)			
SCRUM-82 **Identity** User Login & Session Module Suite 2: Login & Session (`TC-LOG-01..07`)	SCRUM-82		
Identity Password Reset & Recovery Module Suite 3: Recovery (`TC-RST-01..05`)	SCRUM-82		
Identity User Profile Management Module Suite 4: Profiles (`TC-PROF-01..05`)	SCRUM-82		
Marketplace Property Listing Creation Module Suite 5: Listing Creation (`TC-PROP-01..06`)	SCRUM-83		
Marketplace Unit Availability Module Suite 6: Unit Controls (`TC-AVAIL-01..04`)	SCRUM-83		
Discovery Interactive Map Discovery Module Suite 7: Map Spatial (`TC-MAP-01..05`)	SCRUM-84		
Discovery Property Search & Filter Module Suite 8: Filters & Search (`TC-SRCH-01..06`)	SCRUM-84		
Intelligence AI Rental Assistant Module Suite 9: AI Grounding (`TC-AI-01..05`)	SCRUM-85		
Intelligence Inquiry & Viewing Reservation Suite 10: Inquiries (`TC-INQ-01..05`)	SCRUM-85		
Trust Landlord KYC & Verification Suite 11: Trust & KYC (`TC-KYC-01..05`)	SCRUM-86		
Trust Messaging & Contact Module Suite 12: Communication (`TC-MSG-01..04`)	SCRUM-86		
Governance Review & Rating Module Suite 13: Reviews (`TC-REV-01..04`)	SCRUM-87		
Governance Reporting & Scam Flagging Module Suite 14: Moderation Flags (`TC-REP-01..04`)	SCRUM-87		
Governance Admin Management & Moderation Suite 15: Admin Controls (`TC-ADM-01..05`)	SCRUM-87		
Governance Notification Module Suite 16: System Alerts (`TC-NOTIF-01..04`)	SCRUM-87 +-----+-----		

+-----+-----+ `` --- ## 6. Section E: Unclear Items & Clarifications 1. **Exclusion of Payment Processing:** Reconfirmed. The platform facilitates inquiries and in-person viewings; payments occur off-platform directly between parties. 2. **Dedicated `drawio` Files:** 16 separate `drawio` files will be placed inside `docs/flowcharts/`, giving the instructor a distinct diagram for every module.